

Statistical Visualization I — Stat 5550, Section 001

Fall 2026 (2 Credits)

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https://symanzik.github.io/teaching/2026_stat5550/stat5550.html

Office Hours: Tuesday (T) & Thursday (H) 1:30pm – 2:30pm in RWST 416,
Friday (F) 9:00am – 10:00am (via Zoom), and by appointment.

Classes & Rooms:

TH 9:00am – 10:15am, T 9/15 – H 11/19, 2026 (tentatively): LSB 7 (face-to-face).

Teaching Assistant: To be announced via Canvas.

Please visit the course Web page listed above for emergency announcements, e.g., when Canvas is unavailable. Otherwise, visit Canvas frequently for lecture notes, data sets, R code, etc. — in particular if you miss our face-to-face lectures for any reason. All (additional and updated) materials, announcements, discussions, recordings, etc. from Canvas are part of the course materials. Not seeing one of these in time does not serve as an excuse for not getting point deductions for the course. Deadlines may change or unexpected new regulations and requirements may occur. It is your responsibility to make sure to receive all announcements in time.

Detailed Class Schedule:

For a 2-credit course, we need 20 lectures/lecture days (in contrast to 29 or 30 lectures/lecture days for a 3-credit course). Those days are marked as “Lecture 01” to “Lecture 20” in the overview below:

Week	Tuesday	Thursday
1	9/1 No class	9/3: No class
2	9/8: No class	9/10: No class
3	9/15: Lecture 01	9/17: Lecture 02
4	9/22: Lecture 03	9/24: Lecture 04
5	9/29: Lecture 05	10/1: Lecture 06
6	10/6: Lecture 07	10/8: Lecture 08
7	10/13: Lecture 09	10/15: Lecture 10
8	10/20: Lecture 11	10/22: Lecture 12
9	10/27: Lecture 13	10/29: Lecture 14
10	11/3: Lecture 15	11/5: Lecture 16
11	11/10: Lecture 17	11/12: Lecture 18
12	11/17: Lecture 19	11/19: Lecture 20
13	11/24: Backup	11/26: No class
14	12/1: Backup	12/3: Backup
15	12/8: No class	12/10: No class

Note: “No class” means guaranteed no class that day. I have marked the dates in the three last weeks of the semester as “Backup”, e.g., in case we miss lectures because of weather conditions, sickness, or any other reason. If nothing goes wrong, our tentative last lecture date will be on H 11/19/2026. Backup lectures (if any) will be held via Zoom (and not face-to-face). There is no need to stay in Logan after H 11/19/2026 for a backup lecture in this course.

Course Objectives:

Statistical graphics and data visualization are critical elements of modern data analysis and presentation. From initial exploration of a data set to the final presentation of results to the end user, statistical graphics play a vital role in shaping our understanding of our data. Through proper use of graphics, we can make critical discoveries, and communicate them clearly. Conversely, poor use or misuse of graphics can seriously mislead (by accident or design).

The course will address three main questions:

1. Why statistical graphics (and which ones to draw)?
2. How to construct statistical graphics in R?
3. How to distinguish between **good** and **bad** statistical graphics?

This course is **not** an introduction into a single R graphics package. Rather, a variety of R graphics packages will be used, such as `baseR`, `ggplot2`, `lattice`, etc.

The course will be broken down largely by the dimension of the available data, starting with categorical data. One- and two-dimensional quantitative data sets require and allow far different methods than those of more than three dimensions. Towards the end of this course, we will deal with presentation graphics, including a discussion of tools and principles that lead to a clear communication and those that serve only to confuse or mislead.

Even more than most aspects of statistics, graphics and visualization involve art as well as science. In most cases, there are many reasonable approaches. Only an understanding of the options available and the underlying principles will lead to a successful analysis and presentation.

Prerequisites:

STAT 5050 (“Introduction to R”) with a C– or better. Moreover, you should be familiar with a tool such as R Markdown, knitr, or Sweave that allows you to combine text, R code, graphics, and numerical results in high-quality documents. L^AT_EX is a plus, but is not formally required.

Moreover, I expect that you still have “operational” knowledge from your STAT 2000, STAT 3000, or STAT 5100 (or higher) course. “Operational” means that you still recall sufficient details from regression, ANOVA, hypothesis tests, etc. (it is not sufficient that you have taken such a course several years ago and have forgotten almost all details).

IDEA Center Learning Objectives:

Objective 1) Gaining a basic understanding of the subject (e.g., factual knowledge, methods, principles, generalizations, theories).

Objective 4) Developing specific skills, competencies, and points of view needed by professionals in the field most closely related to this course.

Objective 13) Learning appropriate methods for collecting, analyzing, and interpreting numerical information.

Topics: (subject to change)

1. Introduction.
2. Basic Graph Construction and Refinement.
3. Graphs for Categorical Data.
4. Graphs for Univariate Data.
5. Good and Bad Graphs.

The following topics will be discussed in the follow-up course “Statistical Visualization II”, typically offered in the spring semester: graphs for bivariate and trivariate data, graphs for “hypervariate” (high-dimensional) data, color and cognition, statistical maps, interactive and dynamic graphics, web-based graphics, and history of graphics.

Course Format and Lecture Activity Points:

The course will be offered in a face-to-face format, implying that you should attend most of the lectures in person. There will also be lecture recordings via Kaltura that will be posted in the “Media Gallery” in Canvas in the evenings of a lecture day. I will also open a Zoom link through Canvas on lecture days with planned in-class lecture activities so that you can (partially) participate in those activities and earn some credits for your remote participation. However, there will be limitations to what can be captured via the Kaltura recordings and what can be done via the Zoom link. In particular, the Zoom link will only be an audio link with no video and no view of my computer screen (as Kaltura and Zoom compete for the same devices on my laptop, in particular for the camera). So, try to attend the lectures in person whenever possible.

Under this setup, it will be possible to conduct activities during the lectures and award credit for participating in those activities, ideally in person, but alternatively via Zoom and/or the Canvas “Discussions”. You may be asked to answer some short questions during a lecture, do some programming and then submit your code into the Canvas “Discussions”, or fill out a worksheet on paper. Some of these activities are stand-alone while others build on each other during the course of a lecture (and thus the need to be able to participate in

real time via Zoom). Some lectures may have extended sequences with around 10 (short) activities in a lecture.

Points earned through such activities count as Lecture Activity Points (LAPs). These activities typically will not be announced in advance. In fact, some activities may only be released in Canvas shortly before the start of a lecture. However, if there is no new activity visible in the Canvas “Discussions” at the start of a lecture and there is also no open Zoom link for that lecture, it is safe to assume that there will be no LAPs awarded in that lecture.

LAPs will be awarded on a done (answered) / not done (not answered) basis and not on correctness. This should encourage everyone to think about a question or try some coding without the pressure to do things immediately right when trying.

I cannot say in advance at this time how many lectures there will be where LAPs will be awarded, but I would estimate that there will be somewhere around 12 ± 5 lectures with LAPs (out of the 20 lectures we will have this semester). In my previous courses, students could earn about 80 to 100 LAPs. I would predict that the total number of LAPs will be in a similar order of magnitude this semester. Some of the stand-alone activities may be answered prior to a lecture when posted in a Canvas “Discussion”. Some of the activities will remain open until 11:59pm on a lecture day and can still be answered after class. Extended sequences of activities typically cannot be answered prior to a lecture as they may require some of the lecture materials and answers to some of the earlier activities in such a sequence of activities.

Points from LAPs will contribute to 10% of your course grade. LAPs will translate as follows into course points:

% of LAPs Completed	Course Points
$[90, \infty)$	10
$[80, 90)$	9
$[70, 80)$	8
$[60, 70)$	7
$[50, 60)$	6
$[0, 50)$	0

So, you can miss (for private reasons) some lectures where LAPs are provided and still get the full number of course points for LAPs. However, you should not miss too many of the lectures where a large number of LAPs will be provided. Also, make sure that you do not miss any LAPs later in the semester in case you already missed several LAPs early in the semester. Ideally, plan on attending all lectures in person.

In case of an excused absence, e.g., for medical reasons, family emergencies or funerals, court appointments, university-approved travel, etc., please provide some supporting information and your LAP score will be adjusted according to the number of lectures with LAPs you could attend. Private reasons such as travel, most family events (such as weddings), etc. do not count as an excused absence.

Ideally, bring your laptop to all lectures so you can earn LAPs immediately during a lecture period.

Homework Assignments:

There will be 3 HW assignments for this course, roughly one every three weeks. Each

HW assignment will include a value (typically 20–100 points) that it will be scored out of. **HW assignments will contribute to 90% of your course grade.** The value of each HW assignment will be roughly proportional to its importance and the amount of work involved.

You will be allowed to discuss general approaches to questions on the HW assignments with other students, but each student must write and submit their own R code and comments. Any students caught sharing R code or other parts of their homework submissions will fail the class.

Unless otherwise stated on the HW assignment sheet, all homework assignments have to be submitted electronically via Canvas. HW assignments will be posted tentatively in weeks 6, 9, and 12 of the semester and will be tentatively due in weeks 9, 12, and 15 of the semester (see the *Detailed Class Schedule* above how these weeks approximately translate to dates). **You will have about 2 to 3 weeks after the last lecture to finalize and submit the last HW assignment.**

There will be no (in-class or take-home) quizzes, midterm exams, or final exams. Nevertheless, this will be a very challenging course that requires a lot of individual time to work on the assignments. Just attending classes will not be enough to pass this course! **In addition, you will have to do a lot of individual reading of textbooks, online documentation, and help pages, and search for available information on the web.**

The use of Artificial Intelligence (AI) tools such as ChatGPT, Co-Pilot, and others is permitted, but has to be acknowledged in your submissions.

No Excuse Needed Late Homework Submission Policy:

Each student has **3 tokens** for late homework submissions. These tokens can be used in multiple ways, e.g., one token at a time for each of the 3 HWs or all 3 tokens can be used for a single late HW submission. You will need 1 token if your HW is 1 min – 24 hours late; you will need 2 tokens if your HW is > 24 hours – 48 hours late; and you will need all 3 tokens if your HW is > 48 hours – 72 hours late. Once you have used up all 3 tokens, late HWs will count as 0 points, even if such a HW is only one additional minute late. As an example, if you used 2 tokens early on, you will have only 1 token left and can submit your next HW only up to 24 hours late. If you submit that HW more than 24 hours late, it will count as 0 points. All times are based on the Canvas time stamp that is created when a HW gets submitted — and not on your local computer clock.

It is your responsibility to keep track of the number of tokens you have used. Record that information on your phone or check how you can extract the submission dates and times in Canvas on your side. I will not be able to tell you shortly before the next submission deadline how many tokens you have left. In fact, I only plan to use that information when I determine final course grades. You may therefore get assigned a score > 0 points for a certain HW submission initially, but that score may get adjusted to 0 points if it turns out that this was for a late submission and you had already used up all your tokens.

You will not get any credit for any unused tokens, so you could use the remaining ones for proofreading, checking that all figures and R code are included, etc., in particular for the later HWs — and if you still have tokens left at that time!

As the name indicates, there is no excuse needed when you use a token, be it for a minor sickness, personal travel due to family events, or just needing more time as you couldn't

finish that HW by the deadline. If you are officially absent from USU when a HW is due, e.g., due to travel with a USU sports team or band, attending a conference, doing fieldwork, etc., I need an official statement from a coach or supervisor — and no token will be used. However, we then need to determine on a case-by-case basis when that particular HW will be due for you — and when you would have to use additional token(s) to further extend your extended deadline.

You can resubmit your HW as often as you want in Canvas prior to the deadline. Apparently, if you resubmit it after the deadline, one (or more) tokens will be used. In that case, also let me know via e-mail at the time of the deadline that you are still working on your HW beyond the original deadline and plan to resubmit a new version later on. Once a HW has been graded on my side, you will no longer be allowed to resubmit a revision of that HW (or in case you did, you still would get the points based on the last submission prior to the original deadline).

Grading Scheme:

I do not specify any definite grade cutoffs in advance, but I typically assign the same grade to all students that fall into the same narrow cluster of points that extends below one of the commonly used grade cutoffs (i.e., 93% for A, 90% for A–, 87% for B+, 83% for B, 80% for B–, 77% for C+, 73% for C, 70% for C–, 67% for D+, 60% for D, and <60% for F). So, for example, if one student earned 92.9% of points and the next higher student earned 93.1% of points, both students almost certainly will get an A grade. However, if one student earned 92.9% of points and the next higher student earned 95.0% of points, this will almost certainly translate into A– and A grades, respectively.

Textbooks:

Tufte, Edward R. (1983) *The Visual Display of Quantitative Information (2nd Edition)*, Cheshire, CT: Graphics Press,
<https://www.edwardtufte.com/book/the-visual-display-of-quantitative-information/>.

Unwin, Antony (2015) *Graphical Data Analysis with R*, Boca Raton, FL: CRC Press/Taylor & Francis,
<https://www.routledge.com/Graphical-Data-Analysis-with-R/Unwin/p/book/9781498715232>,
and
<http://www.gradaanwr.net/>.

Wickham, Hadley (2016) *ggplot2 — Elegant Graphics for Data Analysis (2nd or 3rd Edition)*, New York, NY: Springer,
<https://www.springer.com/us/book/9783319242750>,
<https://ggplot2-book.org/>,
<https://ggplot2.tidyverse.org/>, and
<https://ggplot2.tidyverse.org/reference/>.

Every student should have access to each of these books, but it is not necessary that every student buys all of these books. The USU library holds several of these books or provides online access. If you plan to work in the area of Visualization and/or Data Science for your MS or PhD degree, you should consider to purchase these books for an ongoing use beyond this course.

Software:

We will primarily be using R (<https://cran.r-project.org/>), a free software environment for statistical computing and graphics. Please install a recent version of R, i.e., 4.6.1

or later, on your own computer so we can exchange code. Working with version 4.6.0 should still be OK. I would discourage the use of version 4.5.3 or older. Also install the most recent version (2026.08.2 Build 200, released on 2026-08-26) of RStudio (<https://posit.co/>) as a front end to R. **Note that some versions of RStudio released earlier in 2024 contained bugs that prevented them from opening certain Rnw files, similar to the ones used in class. If you get error messages when you try to open some of the files from class, check whether you have the most recent version of RStudio installed — and if not, upgrade to the most recent version.** For Windows, also install Rtools 4.5 (<https://cran.r-project.org/bin/windows/Rtools/>).

For most work in this course, using the *tinytex* R package is all that is needed, but if you plan to work on more complicated L^AT_EX documents later on (such as your MS thesis or your PhD dissertation), you should also install a full version of L^AT_EX that can be used in connection with RStudio. My personal recommendation is TeX Live (<https://www.tug.org/texlive/>).

We will not make use of Tableau in this course and we also will not work directly with Python.

Courtesy:

Please turn off cell phones and similar devices before class, and please keep conversations to a minimum during lectures. Please do not read/reply to your e-mails or browse other web pages than the ones discussed during class.

I will not keep track if you come to class or not. However, I would highly recommend to attend all lectures so you can immediately ask questions and earn LAPs when assigned.

Americans with Disabilities Act:

USU welcomes students with disabilities. If you have, or suspect you may have, a physical, mental health, or learning disability that may require accommodations in this course, please contact the Disability Resource Center (DRC) as early in the semester as possible (University Inn Room 101, 435-797-2444, drc@usu.edu, <https://www.usu.edu/drc/>). All disability related accommodations must be approved by the DRC. Once approved, the DRC will coordinate with faculty to provide accommodations.

Note:

The above schedule and procedures in this course are subject to change in the event of extenuating circumstances.